Practice Quiz 07B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 07

Construct a counterexample to preservation of transitivity.

1. Let $A = \{1, 2, 3\}$. Construct two nonempty transitive relations R and S on A such that $R \cup S$ is not transitive. List their ordered pairs, justify that each relation is transitive, and exhibit a failure of transitivity in their union.

   Solution Take $R = \{(1, 2)\}$ and $S = \{(2, 3)\}$. Both are nonempty relations on A.

   For R, there are no u, v, w with both uRv and vRw: the first relation would require $v = 2$, and the second would require $v = 1$. Thus the implication in the definition of transitivity always holds, so R is transitive. The same reasoning applies to S: two successive S-pairs would require $v = 3$ and $v = 2$. Hence S is transitive.

   However, $(1, 2)$ and $(2, 3)$ belong to $R \cup S$, while $(1, 3)$ does not. Therefore $R \cup S$ is not transitive. ∎

Solutions revised using GPT-6 Astra in the Codex harness.